



DATA MINING & WEB ANALYTICS

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1. Introduction

1A: Problem

The UK Used Car Market is witnessing an increasing trend in which the market is showing consistent year-on-year increases in demand, whereas demand for new cars is declining. This has indicated a strong shift in buyer behaviour towards used cars (Business Car, 2025). Hence, there is a need to ensure competitive pricing. The most important factors regarding vehicles in the market are the Mileage, Age and the Fuel Type of the vehicle. Since it is a transparent market where pricing sensitivity is significant, higher prices can be a potential deterrent for customers, whereas if the prices are too low, there will be an adverse effect on profitability. This data-driven approach will not only help to achieve an appropriate pricing balance but will also ensure appropriate pricing decisions are made.

1B: Stakeholders involved and why it matters to them

S. No	Stakeholder	Why It Matters?
1	Car Dealerships	-Able to price their cars competitively -Maximise profits -Optimise inventory and purchase decision
2	Customers	-Better transparency when buying second-hand cars -Fairer pricing and purchase planning -Improved decision making
3	Online Marketplace	-Able to handle large amounts of data -Can provide price predictors to guide customers. (can sell as a premium feature) -Help regulate fairer pricing.
4	Insurance Companies	-Accurate Vehicle Valuation -Risk Assessment -Fraud Prevention.

2. Dataset analysis and visualisations

2A: Type, dimensions and variables in dataset

The dataset used in this study is **cross-sectional**, as it captures information on used cars and their associated characteristics at a single point in time. It contains details such as vehicle specifications, usage attributes and prices with each row representing an individual car listing. The data appears to have been sourced from a used-car marketplace or dealership. The dataset includes multiple numerical and categorical variables that describe different aspects of a vehicle, while car price serves as the target variable. During preprocessing, certain variables were reviewed for redundancy and relevancy to ensure model efficiency and interpretability. Features providing overlapping information were assessed, and only the most informative variables were retained for analysis, resulting in a cleaner and more suitable dataset for predictive modelling.

Due to the practicality of this report, 10 features were selected. An initial random forest model was created with all available features and the feature importances (relative to price) were extracted from the model (done on Python). These variables are summarised below:

No	Feature Name	Feature importance score (based on random forest model)	Data type	Feature meaning
1	Monthly Mileage	0.9364	Numerical – Continuous	Average number of miles driven by the car per month
2	Colour	0.0096	Categorical – Nominal	Colour of the vehicle
3	Cylinder Numbers	0.0058	Categorical – Nominal	Number of cylinders in the engine
4	Owners	0.0049	Categorical – Ordinal	Number of previous owners of the vehicle
5	Engine Size	0.0038	Numerical – Continuous	Size or capacity of the car's engine
6	Insurance	0.0024	Categorical – Nominal	Indicates whether collision insurance coverage is present
7	Age	0.0019	Numerical – Discrete	Age of the car in years
8	Peak RPM	0.0018	Numerical – Continuous	Maximum revolutions per minute of the engine
9	Car Width	0.0018	Numerical – Continuous	Width of the car body
10	Fuel Type	0.0010	Categorical – Nominal	Indicates the type of fuel used in the car

Intuitive Call

Fuel Type exhibited a slightly lower feature importance score than Car Length in the random forest model, but it was selected in place of Car Length based on relevance. From a real-world perspective, fuel type is a more meaningful and interpretable determinant of used car prices as it directly influences running costs, fuel efficiency and buyer preferences. In contrast, car length tends to have a more indirect relationship with price and is often correlated with other vehicle dimensions.

2B: Level of data and verbal presentation of two records

Each row (level) shows details about a particular vehicle. To explain this, 2 rows of data are randomly selected.

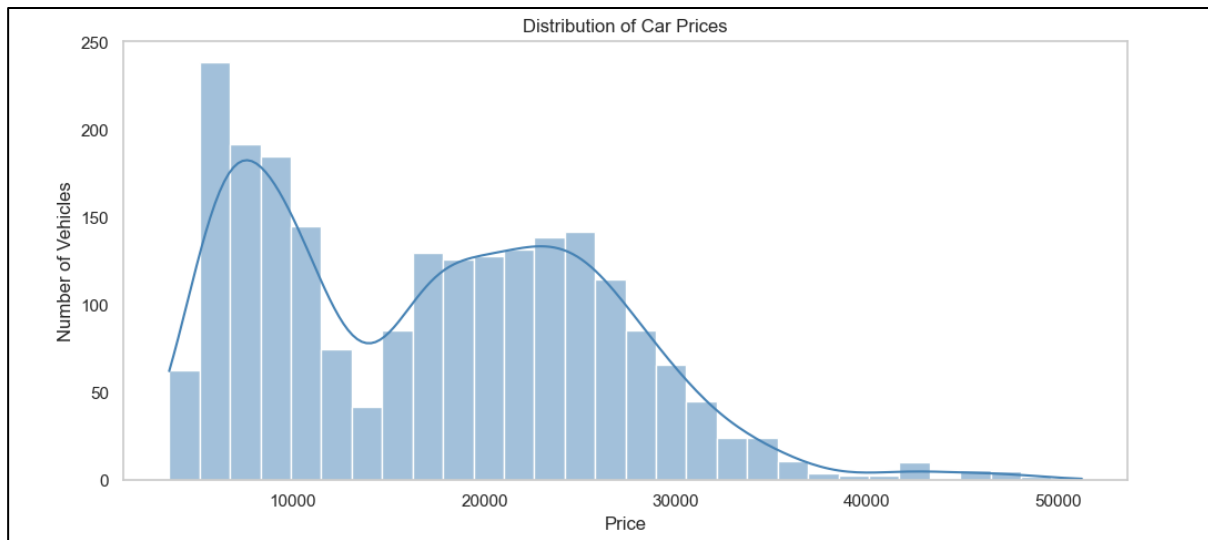
The **43rd entry** of the dataset describes a 2008 yellow Nissan Altima running on petrol fuel with three cylinders. The car has multiple previous owners and includes collision insurance. It has an engine size of approximately 2.40L, a car width of 63.7 units and a peak RPM of 4,437. The vehicle records a relatively high monthly mileage of about 2,750 miles, indicating heavier usage. Based on these characteristics, the car is priced at £9,363.39.

Similarly, the **1605th entry** represents a 2018 black Toyota Corolla with a hybrid fuel type and a five-cylinder engine. The car has fewer previous owners and includes collision

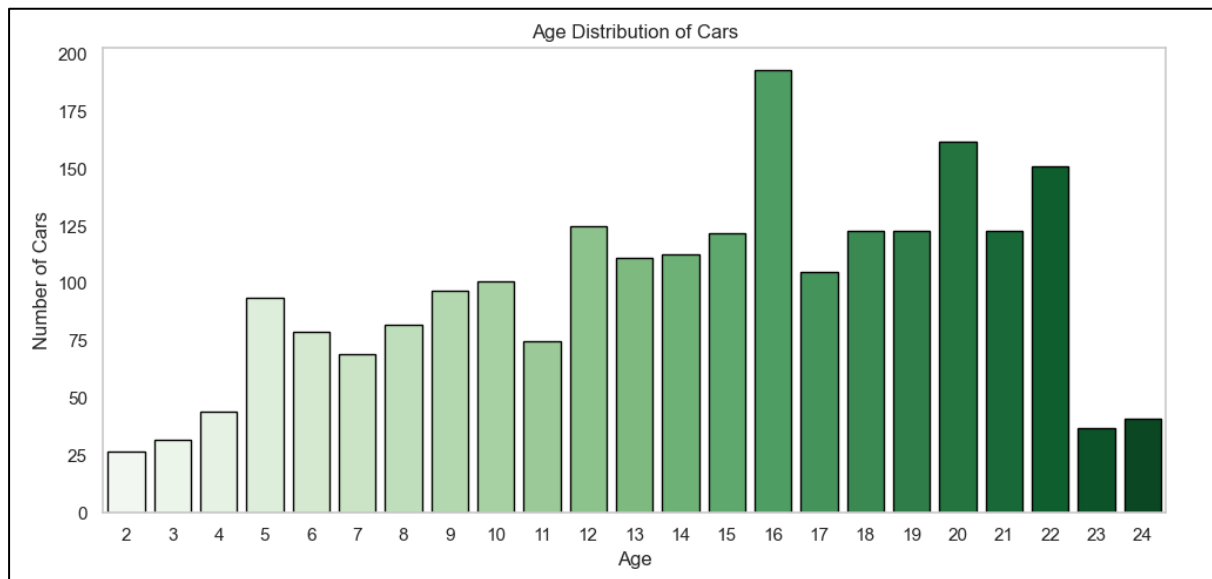
insurance. It is equipped with a 2.52L engine, a wider car width of 71.0 units and a higher peak RPM of 5,416. The monthly mileage is approximately 668 miles, suggesting lighter usage. Reflecting its newer age, lower mileage and stronger specifications, the vehicle is priced at £30,974.98.

2C: Univariate Visualization

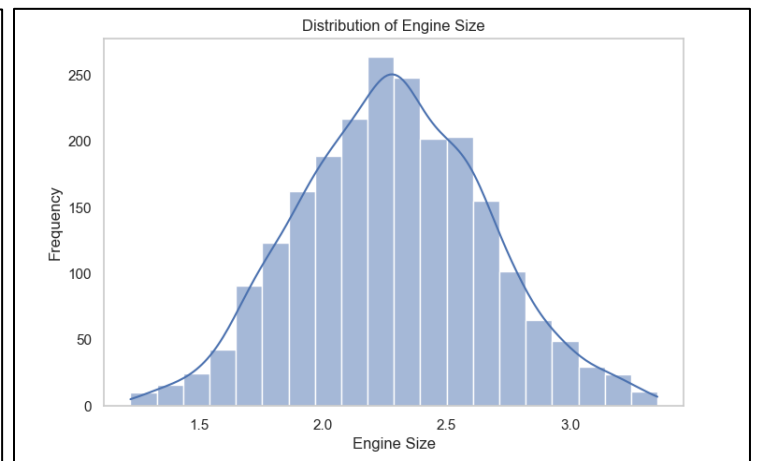
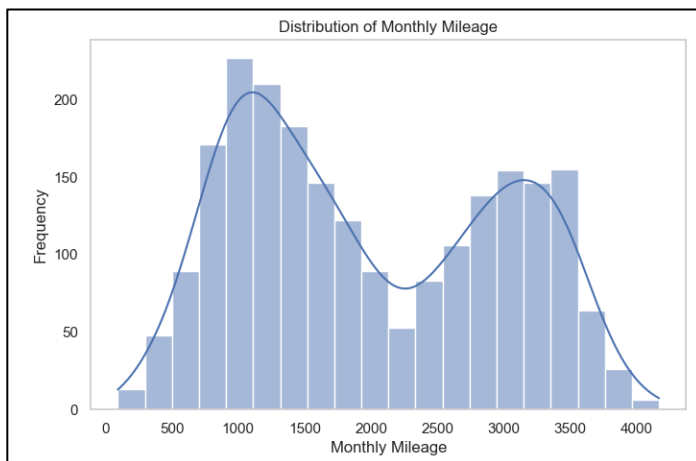
We start with our target variable which is price:



Car prices in this dataset do not follow a normal distribution and instead exhibit two clear peaks. The first peak occurs at approximately **£8,000**, while the second peak is observed around **£22,000–£25,000**, indicating the presence of distinct pricing clusters. In addition, a small number of vehicles are priced above **£30,000**, forming a long right tail in the distribution. This suggests that the used car market captured in the dataset can be broadly divided into **three segments**: lower-priced vehicles, mid-range vehicles, and a smaller high-end segment.



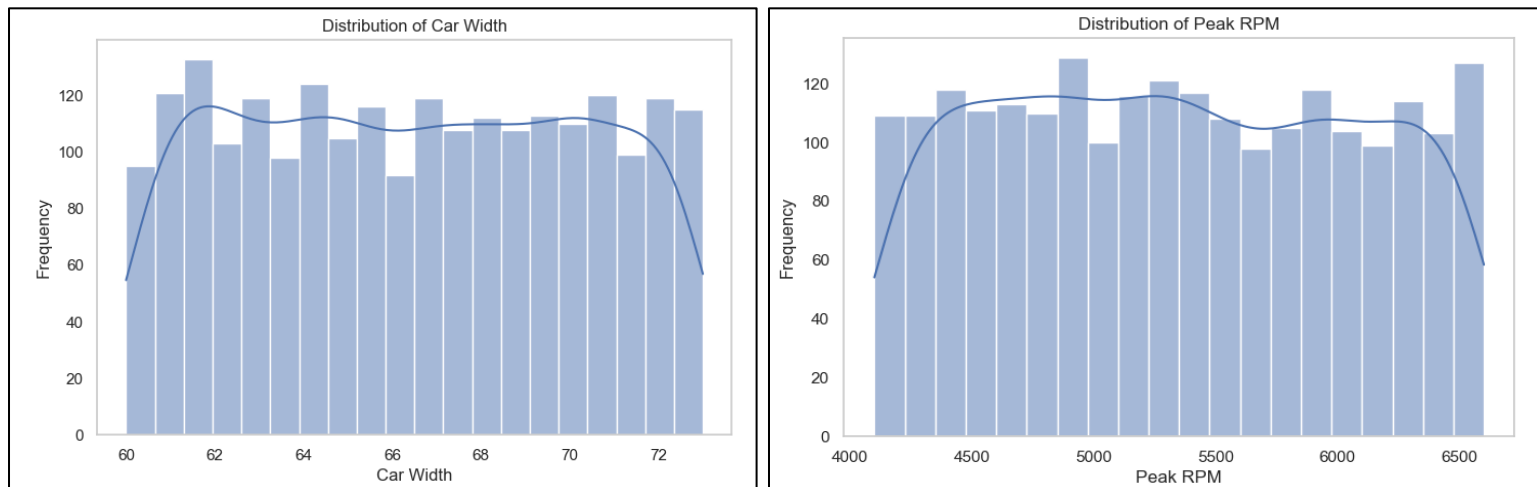
The age distribution of cars in the dataset is **slightly left-skewed**, indicating that a larger proportion of vehicles are older rather than newer. Most cars fall within the **mid-to-higher age range**, with frequencies increasing steadily up to around **15–20 years**, after which the number of vehicles gradually declines. The **mean age is 14.30 years**, with a **SD of 5.67 years**, suggesting substantial variation in vehicle age across the dataset. This indicates that used cars typically enter the resale market after being utilised for a considerable period.



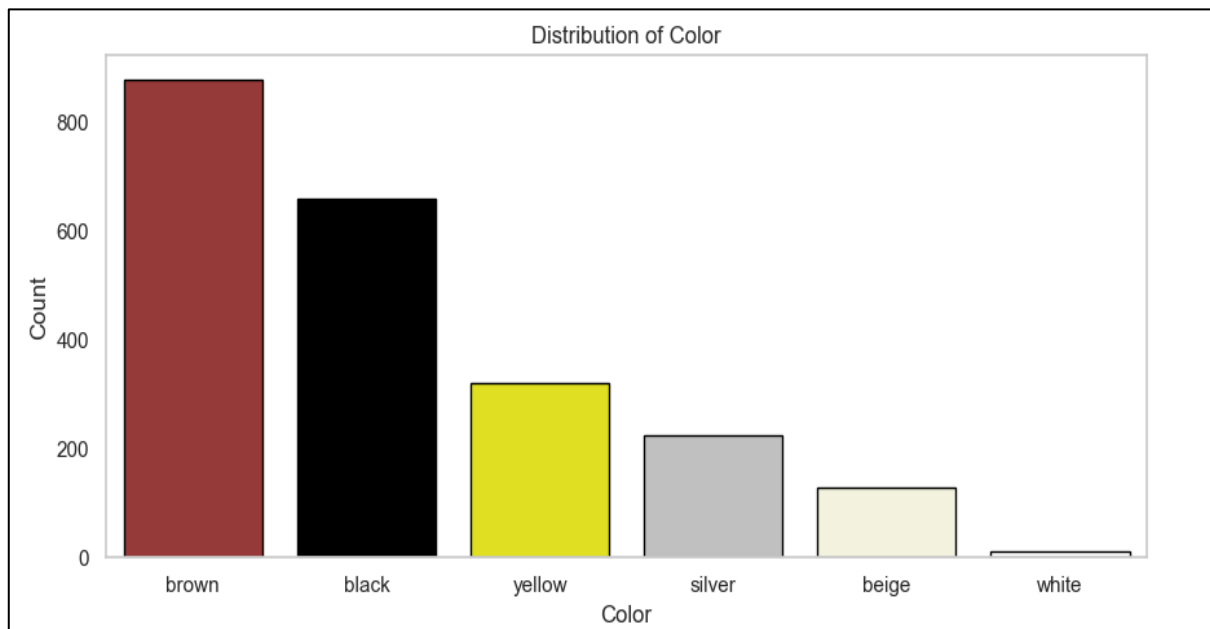
The distribution of monthly mileage is **bimodal**, showing two distinct peaks. This suggests the presence of two driving patterns within the dataset: one group of vehicles with **lower monthly usage** (around 800–1,200 miles) and another group with **higher monthly usage** (around 2,800–3,200 miles). The spread indicates considerable variation in how intensively cars are used which helps explain why mileage is a highly influential predictor of car price.

The engine size distribution is **approximately bell-shaped** with most vehicles clustered between **2.0L and 2.5L**. Very small and very large engines are relatively rare, indicating that mid-sized engines dominate the used car market in this dataset. The limited skewness

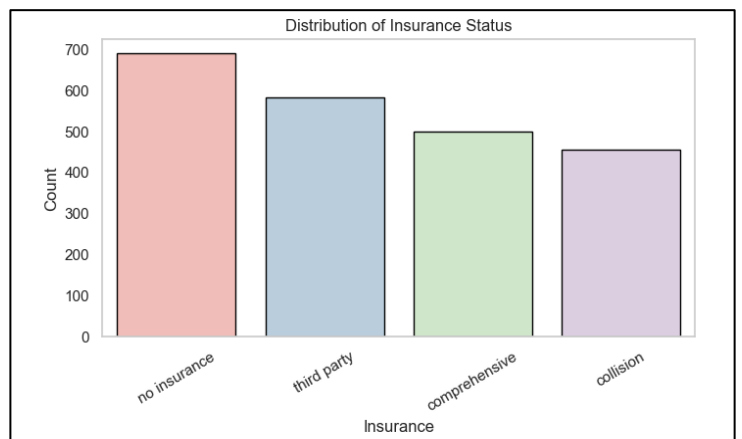
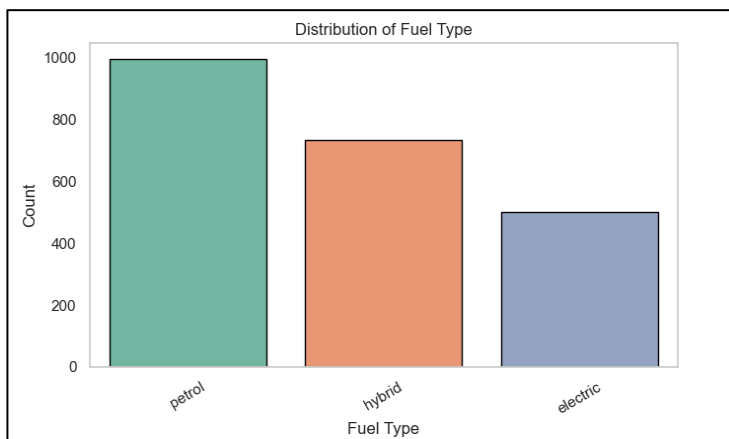
suggests moderate variability making engine size a stable and interpretable feature for price prediction.



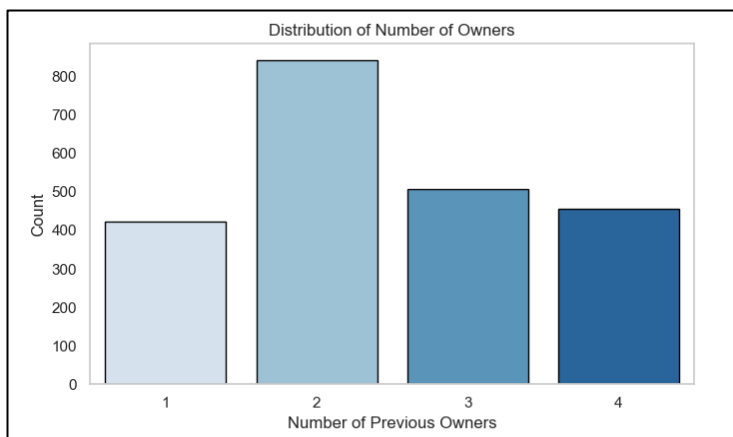
The distribution of car width is **uniform** across the observed range, with most vehicles falling between **approximately 60 and 72 units**. There is no strong skewness or dominant peak, suggesting that car widths are relatively evenly spread within the dataset. Similarly, the peak RPM distribution is also **relatively evenly distributed**, spanning from around **4,000 to 6,600 RPM**. The absence of pronounced peaks indicates that vehicles in the dataset cover a wide range of engine performance levels.



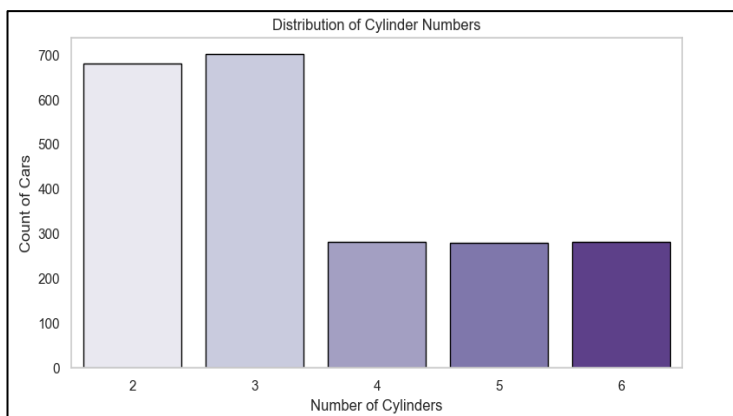
The colour distribution is uneven, with brown and black being the most common colours in the dataset. These are followed by yellow and silver, while beige and white appear far less frequently. However, this does not give any other key information.



The fuel type distribution shows that petrol vehicles make up the largest share of the dataset, followed by hybrid cars. Electric vehicles are the least represented. This reflects current market trends in the used car segment, where petrol cars remain dominant, hybrids are gaining traction, and fully electric vehicles still represent a smaller proportion of listings. For Insurance, there is no clear preferred option for used vehicles and almost 70% of the cars have some sort of insurance out of which majority are insured with a third party.



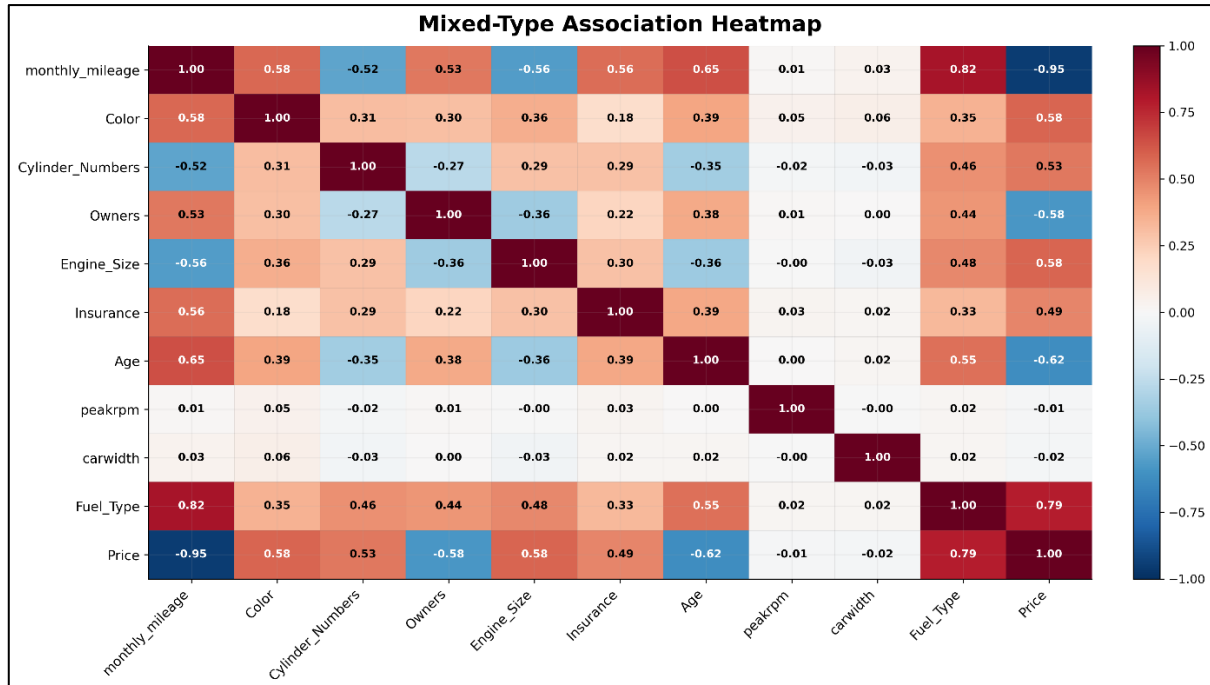
When it comes to previous owners, the distribution indicates that cars with two previous owners are the most common in the dataset. Vehicles with one owner are less frequent while those with three or four previous owners also form a substantial share. This pattern suggests that most used cars change hands multiple times before resale and ownership history is an important factor that may influence buyer perception and pricing.



The distribution of cylinder numbers shows that vehicles with two and three cylinders are the most common in the dataset, indicating a strong presence of smaller and more fuel-efficient engines. Cars with four, five, and six cylinders appear less frequently and are relatively evenly distributed among themselves.

2D: Bivariate Visualization

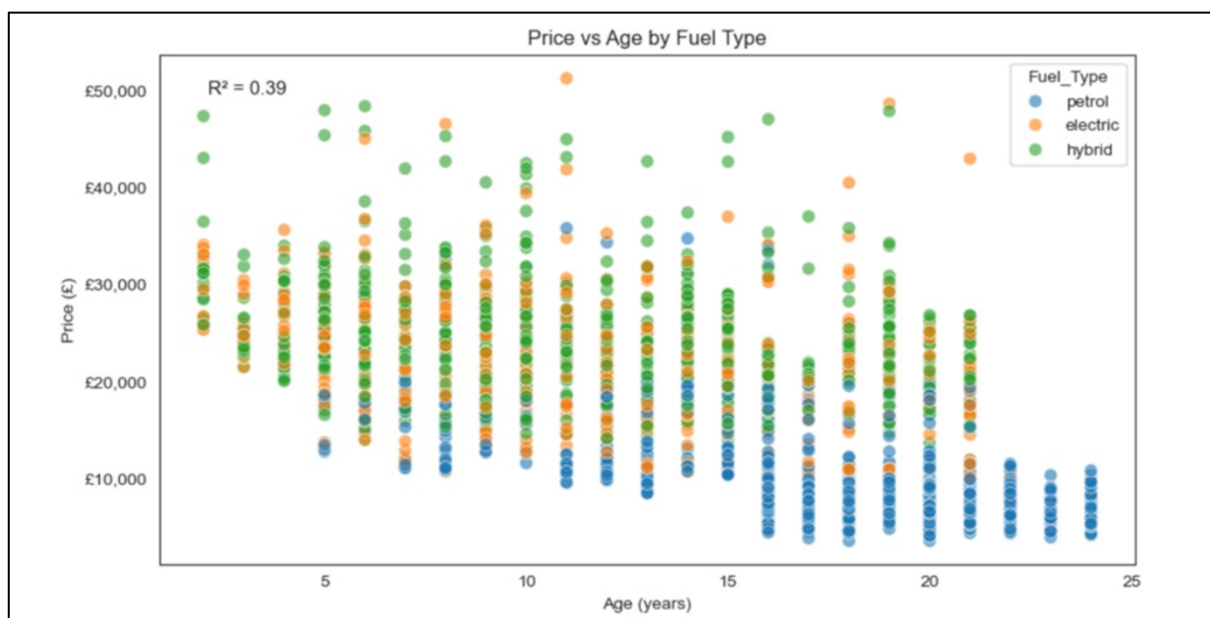
A heatmap was made to determine the relationship between the different variables and price.



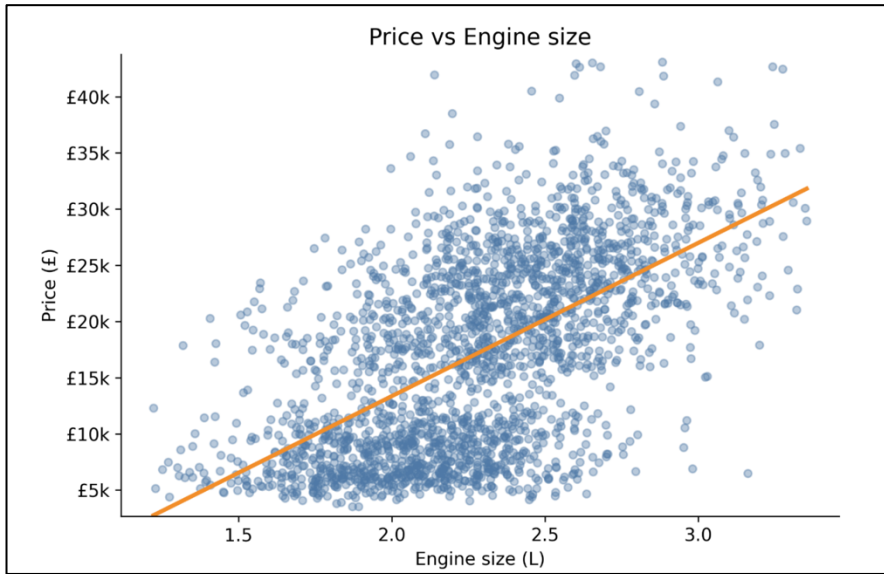
The strongest relationship is between monthly mileage and price (-0.95), indicating a very strong negative association. This suggests that as monthly mileage increases, the price of the car decreases significantly. Mileage appears to be the most influential factor in determining price. There is also a strong positive association between Fuel Type and Price (0.79) which also corroborates the intuitive approach we took earlier for feature selection. Age (-0.62) and Owners (-0.58) show moderate negative associations with price, meaning older cars and cars with more previous owners tend to be priced lower.



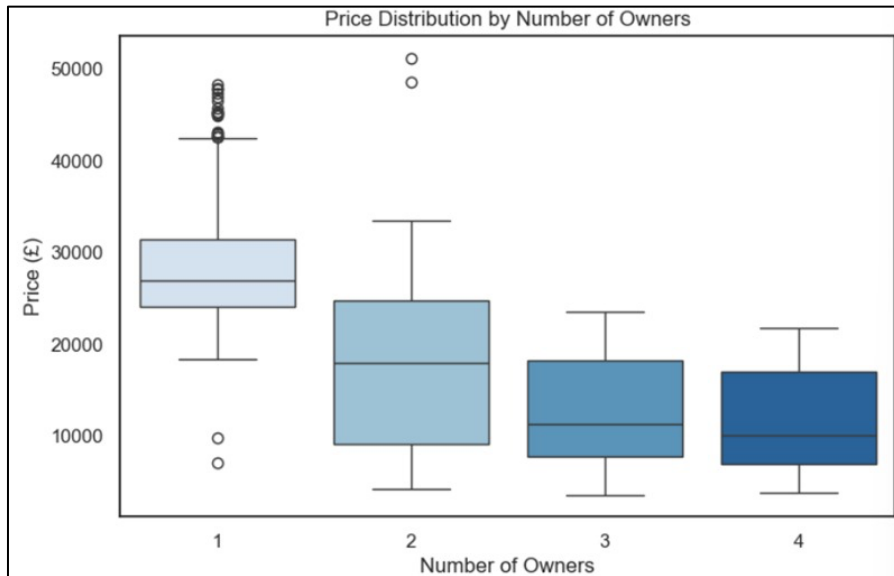
The scatter plot shows a strong negative relationship between monthly mileage and price, confirming that as mileage increases, car prices decrease significantly. Vehicles with lower monthly mileage (below 1,000 miles) are generally priced much higher, often above £30,000, whereas cars with higher mileage (above 3,000 miles) tend to be priced below £10,000.



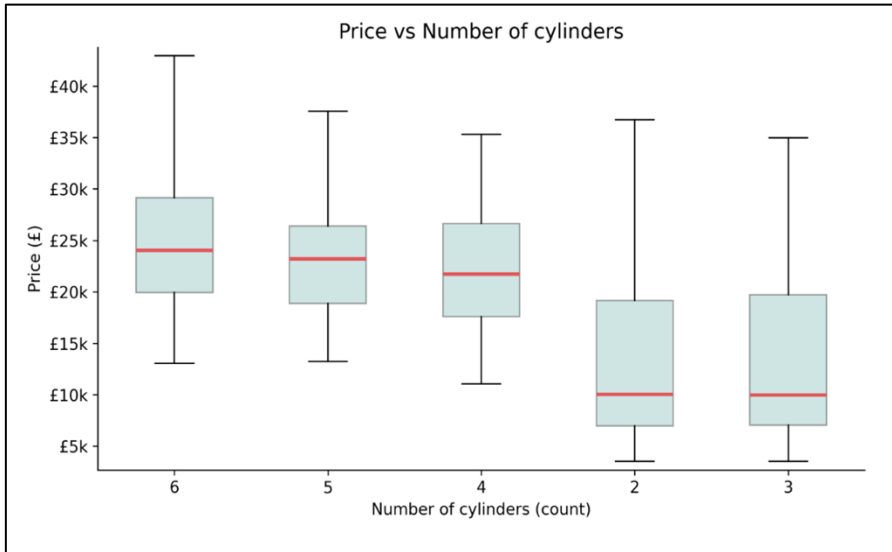
The scatter plot above shows a moderate negative relationship between age and price, indicating that car prices generally decrease as vehicles become older. The reported R^2 value of 0.39 suggests that age explains a meaningful, but not dominant proportion of price variation. When segmented by fuel type, hybrid and electric vehicles consistently appear in higher price ranges, particularly at lower ages. Petrol vehicles are generally priced lower across most age groups and show a clearer downward trend as age increases.



The scatter plot on the left shows a **clear positive relationship between engine size and price**. As engine size increases, the price of the vehicle generally rises, which is confirmed by the upward-sloping regression line. Cars with smaller engines (around 1.4–1.8L) are mostly concentrated in the lower price range, typically below £15,000.

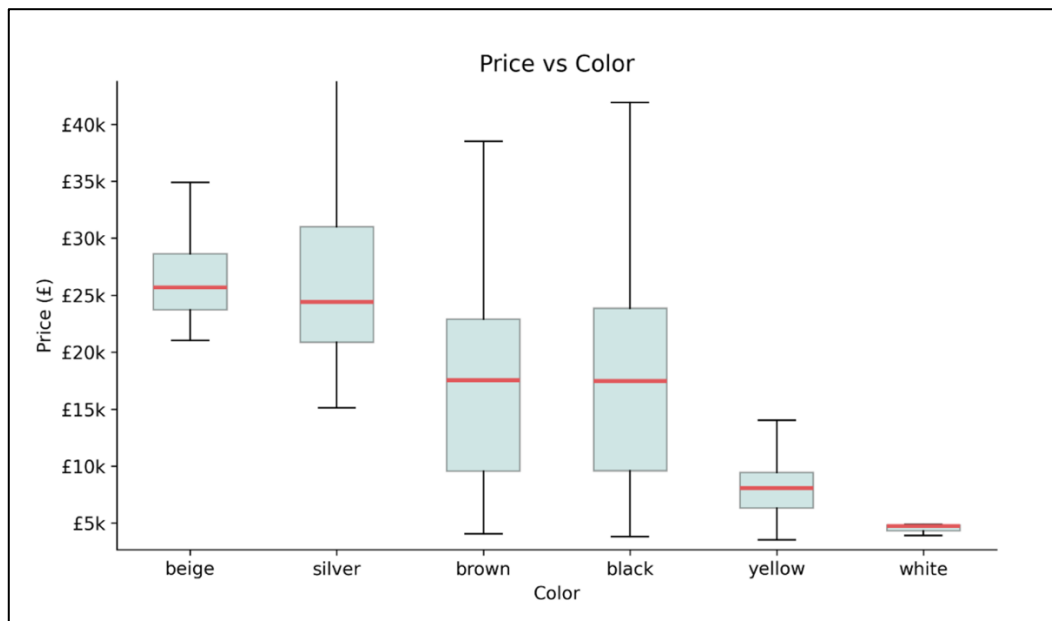


Similarly, the boxplot shows a clear negative relationship between the number of previous owners and price. Vehicles with one previous owner have the highest median price and the widest price range, indicating stronger resale value.



Vehicles with five and six cylinders have the highest median prices, followed by four-cylinder cars. In contrast, cars with two or three cylinders have noticeably lower median prices.

Finally, the boxplot below shows the relationship between Price and Colour. Beige and silver vehicles have the highest median prices, indicating that these colours are generally associated with higher-value cars. Brown and black cars fall into the mid-price range, with moderate variability in prices. In contrast, yellow and white vehicles have the lowest median prices, with white cars showing very limited price variation and consistently lower values.



2E: K – Means Clustering (Unsupervised)

Rationale

K-means clustering was applied to explore whether cars in the dataset naturally form distinct groups based on their characteristics. The aim was not to predict price, but to better understand different types of vehicles in the market.

Optimal Number of Clusters

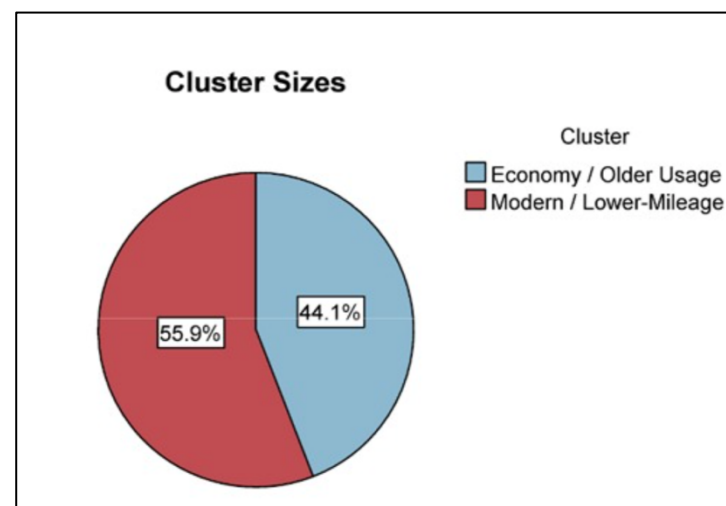
Number of Clusters	Silhouette Score
<u>k</u> =2	0.3
<u>k</u> =3	0.2
<u>k</u> =4	0.2
<u>k</u> =5	0.2
<u>k</u> =6	0.1
<u>k</u> =7	0.1
<u>k</u> =8	0.1
<u>k</u> =9	0.1
<u>k</u> =10	0.1

Several cluster solutions were tested using different values of k. Based on the silhouette score, k = 2 provided the most stable and meaningful result. This indicates that the dataset can be divided into two main vehicle segments.

Cluster Size & Meaning

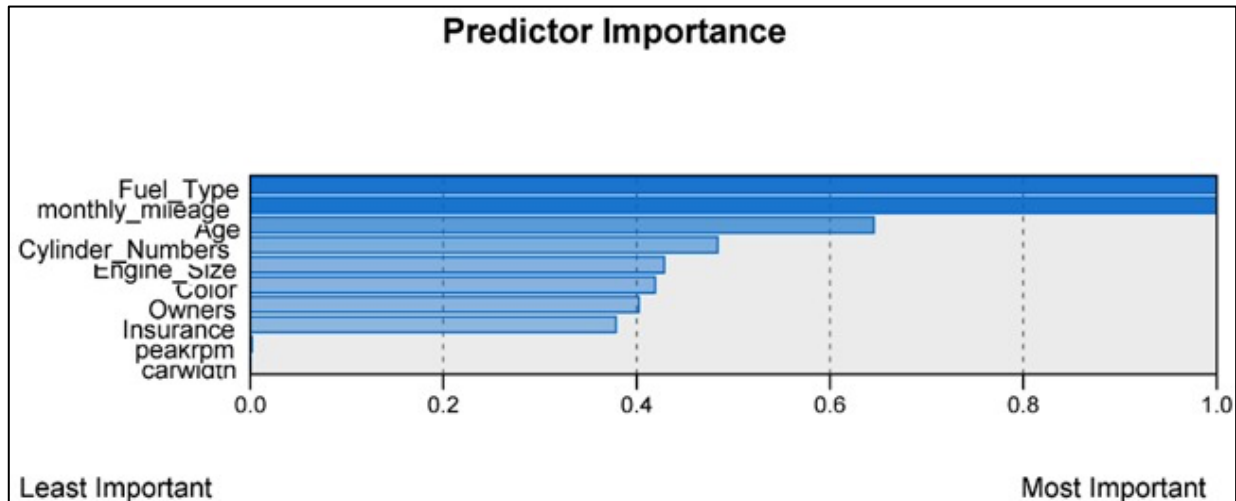
The first cluster contains more modern vehicles with lesser mileage and higher prices. These vehicles are obviously newer and of greater market value. On the other hand, the second cluster includes older vehicles and possibly those with higher mileage, which equate to lower prices. Obviously, this contains the more economical vehicles for the market. There is a significant balance between both groups, with approximately 56% belonging to the modern/low mileage cluster and 44% belonging to their older/economy counterpart. This shows that there is a fair representation of the vehicle market.

Size of Smallest Cluster	983 (44.1%)
Size of Largest Cluster	1246 (55.9%)
Ratio of Sizes: Largest Cluster to Smallest Cluster	1.27



Predictor importance

The predictor importance chart shows which variables contributed most to separating the clusters.



Fuel type appears as the strongest factor, followed by monthly mileage and vehicle age. This confirms that newer, lower-mileage cars form a distinct segment from older and more heavily used vehicles. Engine size, number of cylinders and colour have a moderate influence, while peak RPM and car width contribute very little to cluster separation.

Overall, the clustering highlights that fuel type, mileage and age are the main factors distinguishing cheaper vehicles from more premium ones. These findings are consistent with the earlier exploratory analysis and help provide a clearer understanding of market segments within the used-car dataset.

2F: Data Quality Assessment

Data cleaning

Some values had to be corrected due to possible typos and different notations for blanks, summarised in the table below:

Sr No	Column Name	Original Value	Revised Value
1	Colour	'Brwn'	'Brown'
2	Cylinder Numbers	Some values were strings like 'two', 'three', 'four'	Converted to numbers 2, 3, 4 etc.
3	Insurance	'no' and '3 rd party'	'No insurance' and 'third party'
4	Fuel Type	Spelling correction like 'petroll'	'Petrol'
5	Monthly Mileage	Negative values	Removed them
6	Owners	0 owner entries	Replaced with Mode

Outliers

Outliers in the Engine Size variable were identified using the Interquartile Range (IQR) method. The first quartile (Q1) and third quartile (Q3) were computed and the IQR was defined as $IQR = Q3 - Q1$. Observations falling below $Q1 - 1.5 \times IQR$ or above $Q3 + 1.5 \times IQR$ were classified as outliers. Rather than removing these extreme values, they were replaced with the median engine size calculated from the non-outlier observations. This approach preserves the overall sample size while reducing the influence of extreme values on model estimation. Median imputation was selected because it is robust to skewness and provides a stable central estimate.

Missing Values

While the complete dataset contained many missing values, the number of missing values on the selected columns were as follows:

[23]:		Missing_Count
	Price	0
	monthly_mileage	0
	Owners	0
	Fuel_Type	3
	Color	2
	peakrpm	0
	carwidth	0
	Engine_Size	0
	Age	0
	Insurance	0
	Cylinder_Numbers	0

For Fuel Type and Colour, the missing values were replaced by the most frequent category i.e. Mode.

3. Data Modelling and results

3A: Predictive modelling formulation and problem type

The goal of the model is to predict second-hand car prices based on its features. This is framed as a supervised regression problem, where previously priced cars are used to train the model.

3B: Data Partitioning

Data was split into 80% for training and 20% for testing. The split allows the models to learn patterns from the training set and evaluate its performance on unseen data (testing set). This reduces the risk of overfitting where the model might give better results on familiar data but poorly on unseen data.

3C: Performance Metrics Selection

Mean Absolute Error (MAE) is used to evaluate the model's performance because:

- MAE measures the average absolute difference between the actual and predicted values, providing a direct indication of how far predictions deviate from true prices.
- Unlike MSE, MAE does not square the errors which means it does not disproportionately penalise large deviations. This makes it less sensitive to extreme values and more robust in datasets where price variability is high.

3D: Feature engineering

For modelling purposes, age was computed using the year (i.e. 2024 – 'Year'). Age is more meaningful in the context of used cars and is easier for models to interpret since it starts from 1 and is intuitive, unlike year on its own.

3E: Baseline Model

Since this is a regression problem, the mean is used to set the baseline. The mean was identified on the training set (£17,296) and this single value was used as a prediction for all the test data. The MSE, RMSE and MAE of this were:

[12]:	Data	MSE	RMSE	MAE
0	Training	8.088996e+07	8993.884494	7671.856687
1	Testing	7.658764e+07	8751.436671	7473.504280

Clearly, this is a poor prediction since the error is over £7,000. Subsequent models are evaluated against this to benchmark performance.

3F: K-Nearest Neighbours

K-Nearest Neighbours (KNN) is a distance-based supervised learning model that predicts the target variable based on the average value of the k closest observations in the feature space. The model assumes that similar vehicles (in terms of mileage, engine size, age, etc.) should have similar prices. Using SPSS Modular, different values of k were tested to identify the optimal balance between bias and variance. The best-performing model was obtained at k = 3, which produced a testing MAE of £1,899.27. Further datapoints of this outcome are given in Appendix 1.

Model	Hyperparameters	MAE Training	MAE Testing	Difference in training and testing
KNN	k=3	1722.627	1899.27	176.643
KNN	k=5	1875.797	1966.166	90.369
KNN	k=7	1953.686	2078.105	124.419
KNN	k=9	1979.601	2132.597	152.996
KNN	k=11	2025.611	2151.777	126.166

3G: Decision Tree

Decision Trees are supervised learning models that split the dataset into branches based on feature thresholds, creating a tree-like structure. Each split is chosen to minimise prediction error, allowing the model to capture non-linear relationships between predictors and price. Hyperparameters tuned included maximum depth and minimum samples required for a split. The minimum error of £1496.60 was achieved at depth = 5. This hyperparameter not only gives the lowest difference between training and test MAEs but also keeps the complexity of the model in check with minimum iterations. Further datapoints of this outcome are given in Appendix 2.

Model	Hyperparameters	MAE Training	MAE Testing	Difference in training and testing
Decision Tree	Default (depth=5)	1355.129	1496.604	141.475
Decision Tree	Max depth= 6	1342.249	1502.869	160.62
Decision Tree	Max depth= 7	1327.943	1497.532	169.589
Decision Tree	Max depth= 8	1327.943	1497.532	169.589
Decision Tree	Max depth= 9	1327.943	1497.532	169.589

3H: Random Forest

Random Forest is an ensemble learning method that constructs multiple decision trees using bootstrapped samples and random subsets of features. The final prediction is obtained by averaging the predictions from all trees, which helps reduce variance and improve robustness. The lowest MAE on test set was £2,557.06 which was achieved at 50 number of trees. Further datapoints of this outcome are given in Appendix 3.

Model	Hyperparameters	MAE Training	MAE Testing	Difference in training and testing
Random Tree	Number of Trees= 50 Tree depth= 10 Min child node size= 5	2758.21	2564.061	-194.149
Random Tree	Number of Trees= 50 Tree depth= 15 Min child node size= 5	2753.792	2557.891	-195.901
Random Tree	Number of Trees= 50 Tree depth= 20 Min child node size= 5	2753.755	2557.818	-195.937
Random Tree	Number of Trees= 100 Tree depth= 10 Min child node size= 5	2758.21	2564.061	-194.149
Random Tree	Number of Trees= 100 Tree depth= 15 Min child node size= 5	2753.792	2557.891	-195.901
Random Tree	Number of Trees= 100 Tree depth= 20 Min child node size= 5	2753.755	2557.818	-195.937

3I: Artificial Neural Network (MLP)

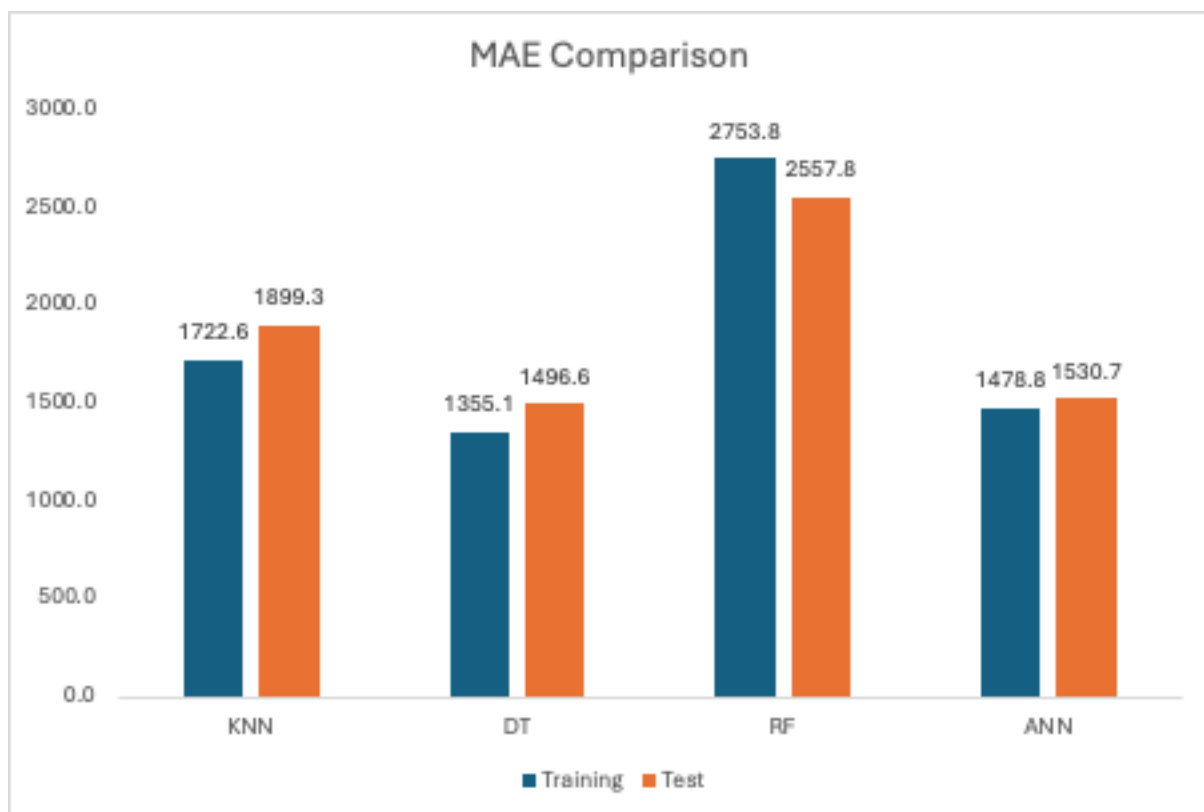
An Artificial Neural Network (ANN) using a Multi-Layer Perceptron (MLP) architecture was implemented to model the nonlinear relationship between vehicle features and price. The best-performing model consisted of a single hidden layer, achieving a testing MAE of £1,530.69. Increasing model complexity by adding a second hidden layer resulted in higher testing error, indicating potential overfitting. Below is the summary of results:

Model Configuration	Hidden Layers	Training Cycles	Training MAE (£)	Testing MAE (£)
NN-1	1	250 (Default)	1478.76	1530.69
NN-1	1	100	1478.77	1530.69
NN-1	1	500	1478.77	1530.69
NN-2	2	250 (Default)	1596.52	1643.79
NN-2	2	100	1596.52	1643.79
NN-2	2	500	1596.52	1643.79

3J: Error Cost Analysis and Summary of Model Performances

The chart below compares the training and testing Mean Absolute Error (MAE) values across the four models. The Decision Tree (DT) achieved the lowest testing MAE £1,496.60, indicating the most accurate average price predictions on unseen data. The Artificial Neural Network (ANN) performed closely, with a testing MAE of £1,530.7, showing good generalisation capability. The K-Nearest Neighbours (KNN) model exhibited higher prediction error £1,899.3, while the Random Forest (RF) model showed the highest testing MAE £2,557.8, despite strong training performance. The noticeable gap between training and testing MAE for RF suggests potential overfitting.

Overall, based on MAE, the Decision Tree provides the best balance between predictive accuracy and generalisation performance.



Error Cost Analysis

The final selected model which is the Decision Tree has an average price error of around £1,497. It is insignificant considering the price range of second-hand cars in the UK market. Therefore, this model is reliable and viable in the industry.

For car dealerships or sellers pricing too low means that the seller is losing money on each sale. Pricing too high means that cars are left unsold for longer, resulting in higher storage costs. The model helps reduce both risks by suggesting balanced and competitive prices.

For the customer, a more accurate pricing structure means greater transparency. When prices reflect the actual market value, buyers feel more confident in placing their trust and purchase decisions become easier.

For online shopping platforms, better predictions of prices would help in gaining customer trust and provide reliable price guidance tools.

For insurance companies, vehicle valuation helps them compute insurance premiums accurately. This minimizes the chances of people being either under-insured or over-insured.

4. Conclusion

This analytical study developed predictive models to estimate used car prices based on key vehicle features. After thorough data cleaning, univariate and bivariate analysis, multiple models were trained and evaluated. The Decision Tree (DT) model seemed to have the best predictive power to predict second hand car prices.

Monthly Mileage, Fuel Type and Age are the most influential factors in determining car prices and hence must be considered first when pricing cars for customers whereas other features could be negotiated on.

This model can assist sellers in setting car prices competitively so they can be cheaper than other sellers while also maintaining profitability. The model can be improved by:

- Incorporating additional data such as regional demand patterns, brand reputation and seasonal trends could refine accuracy
- Updating the model regularly with new sales data would ensure it remains aligned with changing market conditions. More training data will allow better price predictions to be made.

REFERENCES

Business Car (2025) *UK used car sales growth reported*. Available at: <https://www.businesscar.co.uk/news/uk-used-car-sales-growth-reported/> (Accessed: 12 February 2026).

APPENDICES

APPENDIX 1

KNN		
'Partition'	1_Training	2_Testing
Minimum Error	-11470.661	-7299.368
Maximum Error	20345.726	15895.496
Mean Error	14.093	311.461
Mean Absolute Error	1722.627	1899.27
Standard Deviation	2433.692	2676.886
Linear Correlation	0.962	0.954
Occurrences	1,777	452

APPENDIX 2

Decision Tree		
'Partition'	1_Training	2_Testing
Minimum Error	-12620.942	-11425.227
Maximum Error	10670.187	10601.246
Mean Error	-28.974	62.612
Mean Absolute Error	1355.129	1496.604
Standard Deviation	1985.871	2190.897
Linear Correlation	0.975	0.969
Occurrences	1,777	452

APPENDIX 3

Random Forest		
'Partition'	1_Training	2_Testing
Minimum Error	-8563.783	-7757.015
Maximum Error	25220.097	27827.898
Mean Error	-0.167	38.234
Mean Absolute Error	2753.755	2557.818
Standard Deviation	4159.417	3893.118
Linear Correlation	0.903	0.915
Occurrences	1,777	452